EA Credit: Renewable Energy

This prerequisite applies to

- BD+C: New Construction (1-5 points)
- BD+C: Core & Shell (1-5 points)
- BD+C: Schools (1-5 points)
- BD+C: Retail (1-5 points)
- BD+C: Data Centers (1-5 points)
- BD+C: Warehouses & Distribution Centers (1-5 points)
- BD+C: Hospitality (1-5 points)
- BD+C: Healthcare (1-5 points)

INTENT

To reduce the environmental and economic harms associated with fossil fuel energy and reduce greenhouse gas emissions by increasing the supply of renewable energy projects and foster a just transition to a green economy.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Use on-site renewable energy systems or procure renewable energy from offsite sources for all or a portion of the building's annual energy use.

Choose one or more strategies for renewable energy procurement from the categories below. Points achieved in each category may be added for a total of 5 points.

- Tier 1: On-site renewable energy generation
    - On-site renewable energy generation, environmental attributes (e.g. RECs) retained
- Tier 2: New off-site renewable energy
    - Off-site renewable electricity that is produced by a generation asset(s) built within the last five years or contracted to be operational within two years of building occupancy.
    - Green-e Energy certification or equivalent is required for one-time purchase and delivery of EACs of more than 100% of the project's annual electricity use.
- Tier 3: Off-site renewable energy
    - Off-site renewable electricity that is Green-e Energy certified or equivalent or captured bio-methane

Ownership of Environmental Attributes: All environmental attributes associated with renewable energy generation must be retired on behalf of the LEED project in order for the renewable energy procurement to contribute to credit achievement.

The default contract length for renewable energy procurement is 10 years. Contract lengths less than 10 years may be pro-rated.

For Tier 2, the age of the generation asset(s) is assessed at the beginning of the contract, and the generation asset(s) retain this attribute for the duration of the initial contract or lease term.

Off-site renewable energy must be generated by renewable electricity projects located in the same country or region where the LEED project is located. Methane capture in the form of biogas that is both captured and used on-site may qualify as a Tier 1 renewable energy resource.

Points are awarded according to Table 1, based on the percentage of total site energy use. Renewable electricity and EAC procurement can only be applied to project electricity use or district energy use. Captured bio-methane can only be applied to project fuel use.

Table 1. Points for Renewable Energy Procurement

Points	Tier 1		Tier 2		Tier 3	
	BD+C (Except CS)	CS	BD+C (Except CS)	CS	BD+C (Except CS)	CS
1	2%	1%	10%	3%	35%	10%
2	5%	3%	20%	6%	70%	20%
3	10%	5%	30%	9%	100%	30%
4	15%	10%	40%	12%		
5	20%	15%	50%	15%	-	-

GUIDANCE

Behind the Intent

Renewable energy criteria in v4.1 has been expanded to recognize the variety of procurement strategies that help to add renewable energy to the grid. Renewable energy generation can contribute to greenhouse gas emission reductions and offer local environmental benefits by reducing air pollution and increasing resilience. Renewable energy produced on-site protects projects from energy price volatility while reducing wasted energy lost in transmission.

Additionally, the voluntary market can be an effective catalyst for encouraging energy generators and utility companies to develop clean energy sources and help address climate change. Purchasing off-site renewable energy allows buildings that use nonrenewable power to create market demand for renewable energy and support the development of renewable infrastructure.

Beta Update

This new credit combines Green Power and Renewable Energy Production into one credit, recognizing the wide spectrum of renewable energy procurement. The credit adds new categories of renewables and updates performance requirements. The credit structure incentivizes self-supply of renewable energy and development of new renewables and provides further opportunities for building and portfolio owners to select the renewable procurement strategies that are most appropriate for the project application.

Step-by-Step Guidance

Step 1. Explore opportunities for renewables procurement.
(See Further Explanation, *Renewable Resource Considerations*)

- Step 2. Compare requirements for renewable energy systems and off-site methods of procurement.
  Carefully evaluate the space requirements (for on-site systems), costs, financial incentives, and efficiencies for each potential renewable technology or contract.

- Local funding, financing, and incentives for renewable generation projects may be available for certain technologies and may be a significant factor. When considering funding options, ensure that the terms of the contract will address all renewable attributes to be retained by the project.
- For on-site systems, excess energy, beyond the building's energy demand at a given point, can be sold to the utility company (net metering). The building owner receives the market rate, however, and cannot charge a premium for the renewable energy. In effect, the grid serves as a storage system and frees the project from hosting a storage system on site. Alternatively, project teams may consider including a storage system to increase resiliency, facilitate further control of building energy costs, and reduce renewable energy curtailment.
- Tying into an existing community system or creating a community system may lower cost barriers through economies of scale, because unit costs may decrease as system sizes increase. Community systems can also take advantage of time-shifted demand: one building that is occupied during the day and another building that is occupied at night could both take advantage of the same biofuel-fired heating system. Ensure that the terms of the community renewable energy contract will address all renewable attributes to be retained by the project.
- For buildings that are part of a portfolio of buildings, renewable energy may be available from a third-party system, or the project team may enter an arrangement in which a third party owns a system that serves the project. In such cases, project teams must take additional steps to ensure that the arrangement continues for the contract period required in the credit, and that the renewable attributes are retained throughout the duration of the contract.
- Undertake a cost-benefit analysis to understand the financial and environmental benefit of all available options.
- When considering any financing mechanism or contract for renewable energy procurement, project teams must ensure that renewable attributes associated with the renewable energy generation will be retained by the project (See Further Explanation, *Energy Attribute Certificates*).
- Consider the additional criteria required to demonstrate EAp Minimum Energy Performance and/or EAc Optimize Energy Performance credit for Tier 1 or Tier 2 renewable energy using the ASHRAE 90.1 Appendix G Performance Rating Method or approved equivalent standard. Note that some Tier 1 and Tier 2 systems do not qualify for additional credit under EAp Minimum Energy Performance and/or EAc Optimize Energy Performance (See EAp Minimum Energy Performance, *Further Explanation, Applying Renewable Energy Savings*):

Step 3. Set target for renewable energy procurement.

Select one or more procurement strategies, for a total of up to 5 points. Each procurement strategy must meet or exceed the minimum target based on the percentage of total site energy use specified in the credit language.

To establish the target renewable energy system size for the project, estimate the annual site energy use for the project (See Further Explanation, Total Site Energy Use).

Review credit point thresholds and establish the renewable procurement goals for the project.

Step 4. Finalize Renewable Energy Procurement.

Purchase and install the renewable energy systems, and/or finalize the contract to procure renewable energy or Energy Attribute Certificates (EACs). Assure that the contract is signed by both parties, and that the contract terms confirm all credit requirements.

For on-site renewable energy system(s), review the contract to confirm that the renewable system(s) are scheduled to be operational at the time of building occupancy. On-site renewable systems must also be commissioned per the requirements of LEED EA prerequisite Fundamental Commissioning and Verification and EA credit Enhanced Commissioning, as applicable.

For Tier 2 renewable energy procurement, review the contract to confirm that the renewable system(s) are scheduled to be operational within two years of building occupancy.

Further Explanation

Calculations

Percentage of renewable energy purchased

Use Equation 1 to determine the percentage of renewable energy generated or purchased. Calculate the percentage of renewable energy generated or purchased for each Tier.

Equation 1: Percentage of energy generated or purchased

> % energy generated or purchased = (annual quantity of renewable energy in kWh) / (annual building site energy use in kWh)

Use Equation 2 to estimate annual energy use offset by captured biomethane:

Equation 2: Annual energy usage offset by captured bio-methane

> % fuel energy use offset = (annual quantity of captured bio-methane procured in kBtu) / (annual building site energy use in kBtu)

Use Equation 3 to calculate the number of points achieved for Tier 2 and Tier 3 when the contract term is different than ten years, and/or when procuring renewable energy from multiple Tiers.

Equation 3: Proration of Off-site renewables

Prorated Annual Renewable Percent = (Contract Term (in years)/ 10 years) x Annual Renewable Energy Generation / Annual Building Site Energy Consumption

Where:

- Annual Renewable Energy Generation and Annual Building Energy Consumption use consistent units of site energy (e.g. kWh/year)
- For a one-time purchase, contract term is listed as "one year", and annual renewable energy generation is reported as the total renewable energy generation.

Total Site Energy Use

Total site energy includes all electricity, fossil fuel, district thermal energy, and other site energy delivered to the project.

Projects complying with EA prerequisite Minimum Energy Performance using energy simulation (i.e. Normative Appendix G Performance Rating Method, Section 11 Energy Cost Budget Method, or approved equivalent standard) must use the modeled total site energy documented for the Proposed design before excluding renewable energy as the basis of the renewable energy credit calculations (See Further Explanation, Example 1).

Projects complying with EA prerequisite Minimum Energy Performance through a prescriptive path must use the most recent building Energy Use Intensity by Property Type data from the ENERGY STAR Portfolio Manager Technical Reference to calculate total annual site energy for the renewable energy credit calculations. The site energy must be broken down into electric and non-electric consumption. This is calculated using the total source energy and total site energy reported in the ENERGY STAR Portfolio Manager Technical Reference, and the US source-to-site ratios for electricity and natural gas. If the project is all-electric with no combustion sources for heating, water heating, cooking, etc. in the project and base building, the total site EUI value may be considered as electricity. This energy source calculation is included the LEED v4.1 Renewable Energy Calculator (See Further Explanation, Example 2).

Sum of points from multiple procurement sources

Using Table 1, add up the applicable points from equation 1, for on-site and off-site renewable energy; and equation 2 for captured bio-methane as applicable, for a total not to exceed 5 points.

This summation is performed in the LEED v4.1 Renewable Energy Calculator. For points documented in Tier 2 and Tier 3, total annual electricity procurement is limited to the total proposed annual site energy consumption from electricity and district thermal energy, excluding any Tier 1 electric generation.

Renewable Resource Considerations

The renewable energy credit seeks to increase overall demand for renewable energy and the use of grid-source renewable energy projects, with the goal of supporting broader grid-scale decarbonization. Criteria rewards renewable energy investments that have a high probability of causality (i.e. support development and installation of new renewables) and demonstrate long-term commitment. Project teams should follow a hierarchy for selecting renewable energy sources to meet credit requirements:

- First, on-site generation;
- second, local generation, such as community solar or wind, in instances where it will have a beneficial decarbonizing impact;
- third, offsite generation projects with high probability of causality, e.g. power purchase agreements;
- fourth, renewables from an existing renewable energy project, e.g. utility green tariff or direct access to wholesale markets
- last, energy attribute certificates (EACs)

The U.S. EPA's Guide to Purchasing Green Power provides additional information on the process of and strategies for procuring renewable energy.

Tier 1 On-Site Renewable Energy System Considerations

On-site renewable energy generation, when combined with careful consideration of building energy time of use and grid peak demand, and storage in some grid regions, can reduce annual greenhouse gas emissions, increase building resilience, and support effective grid management.

Eligible On-Site Renewable Energy Systems

To qualify as an on-site system, the renewable energy must be generated on-site from renewable sources produced at the building or contiguous campus site.

Allowable sources for on-site renewable energy include the following:

- Photovoltaic
- Solar thermal
- Wind
- Biofuel harvested on-site and used on-site for thermal and/or electric generation, excluding:
  - Combustion of municipal solid waste
  - Forest biomass waste, with the exception of the following harvested on site: clean urban wood waste, invasive species, habitat restoration programs, clean industrial wood waste (pallets), and tree tops left over from logging operations.
  - Wood coated with paints, plastics, or laminate

- Wood treated for preservation with materials containing halogens, chlorine compounds, halide compounds, chromated copper arsenate, or arsenic; if more than 1% of the wood fuel has been treated with these compounds, the energy system is ineligible
- Geothermal energy, such as electricity or heat generated from subterranean steam or hot water, excluding any geothermal energy used in conjunction with a vapor compression cycle.

Examples of on-site renewable energy generation include:

- A photovoltaic array located on the project site.
- A wind tower located on a contiguous campus owned by the same entity as the project building.
- Landfill gas processed in digesters on a contiguous campus owned by the same entity as the project building and used to produce thermal energy in the project building.

Note: earlier versions of LEED allowed some biofuels produced off-site to qualify as on-site renewable energy. However, based on the clarifications provided in ASHRAE 90.1-2016 for on-site renewable energy, and the clearer distinction between on-site and off-site renewable energy in LEED v4.1, biofuels are only considered on-site renewable systems when the renewable source is harvested on site, and used for on-site generation of electric or thermal energy.

Only usable energy generated from the on-site renewable system shall be considered towards the on-site renewable energy contribution. Usable energy is defined as the output energy from the system less any transmission and conversion losses, such as standby heat loss, losses when converting electricity from DC to AC, or waste heat in a cogeneration system that is exhausted to the atmosphere during periods of low thermal demand. Excess energy, beyond the building's energy demand at a given point, can be sold to the utility company (net metering) when Energy Attribute Certificates (EACs are retained by the project). Net metered electricity may count toward EA Renewable Energy credit up to 100% of annual electricity and district energy use.

A project team should use web resources and other tools available to determine the feasibility of renewable systems, given the project site's climate, context, and infrastructure. Consider the features of the site, such as solar availability, wind patterns, and other renewable energy sources, and any seasonal or daily variations in its supply. Certain project types may have special opportunities: office or university campuses typically have available land, for example, and warehouse projects may have large roof areas.

Match the project's energy needs with renewable energy output when selecting a renewable system. For example, a sunny site is a good candidate for solar thermal hot water, but this type of renewable resource is most cost-effective if the building has a constant demand for hot water. Accordingly, a hotel or a multifamily project may be a better match for a solar thermal hot water system than an office complex.

Daily and seasonal variations in loads also factor into the investigation of renewable energy. For example, a residential project with low daytime electricity demand may require battery storage to benefit from a photovoltaic (PV) array; an office building with high daytime demand may not.

On-site renewable systems must be installed and commissioned prior to the final LEED construction phase project submission to qualify for on-site renewable energy generation credit.

Tier 2 New Off-Site Renewables

Tier 2 off-site renewables are defined as those that have come online within the last five years or are contracted to be operational within two years of building occupancy.

Age of the generation asset(s) is assessed at the beginning of the contract, and the generation asset(s) retain these attributes for the duration of the contract or lease term. Therefore, a contract initiated for a new renewable asset for a portfolio of projects with EACs assigned to the project building would be considered "New renewable" even if the contract period initiated more than five years prior to project occupancy. EACs assigned to the project shall be from the remaining contract duration on or after the year of project occupancy.

Community renewable energy cooperatives, larger-scale investments, such as direct, voluntary purchases in the form of power purchase agreements (PPAs), virtual PPAs, or renewable energy investment trusts, qualify as new off-site renewables provided documentation demonstrates that they meet the criteria described above. Contracts for these investments must indicate the specific system used to generate the renewable energy, with sufficient information available to confirm the renewable system generation capacity and allocation of the EACs (see *Further Explanation, Renewable Attributes*). If the LEED project contracts for a one-time purchase or short-term contract less than ten years for EACs exceeding 100% of the project's annual electricity and district thermal site energy use, the EACs must be Green-e Energy certified or equivalent.

Investment in new off-site renewables creates new renewable energy supply and has the potential to displace energy and emissions from fossil fuel-powered generators, particularly in regions where the grid mix is a higher percentage of fossil fuels.

To qualify as a new renewable system, the contract length shall be a minimum of ten years, or the annual renewable energy generation shall be prorated based on the contract term length. A commitment to renew does not qualify as a new renewable resource.

Tier 3 Off-Site Renewables

Tier 3 off-site renewables are defined as those contracted from an existing renewable energy provider or off-site renewable systems that were contracted for the building after the renewable system came online and came online more than five years before building occupancy.

Existing off-site renewables, which may include utility green tariff programs or direct access to wholesale markets, may be more widely available depending on project location or budget. Investment in existing renewable resources and utility programs remains an important strategy for sustaining market demand for renewables and ensuring financial viability of existing projects.

The contract length shall be a minimum of ten years, or the annual energy renewable energy generation shall be prorated based on the contract term length. Alternatively, Tier 3 procurement only, project teams may show compliance with the ten year minimum contract term by demonstrating the following:

- The project has an executed contract for a minimum of one year, OR where contracts are not available per regulatory requirements, document that the project has been enrolled in the Green-e or equivalent utility tariff for a minimum of one month.
  AND
- The building owner must provide a signed letter of commitment indicating that the project will remain continuously enrolled in the 100% renewable Green-e or equivalent utility tariff, or alternate Green-e or equivalent procurement source for a minimum of ten years (or the number of years documented for credit if less than 10 years).

Energy Attribute Certificates

An Energy Attribute Certificate (EAC) is a transferrable certificate, record or guarantee that provides information about the environmental attributes of one megawatt hour (MWh) of electricity. Renewable generators can sell EACs together with the electricity (bundled), or separately from the electricity (unbundled). EACs are also referred to as Renewable Energy Certificates (RECs) or Guarantees of Origin (GOs).

Retirement of environmental attributes of renewable electricity generation is substantiated through EACs. The owner of the building seeking LEED certification shall contract for renewable energy and demonstrate that the EACs are retained, owned, or retired on behalf of the party who has financial or operational control over the building's electricity use. Note that while the previous versions of LEED allowed EAC arbitrage for on-site systems, LEED v4.1 requires the owner to retain the environmental attributes of renewable energy that is generated on-site in order to for the generation to qualify for credit.

Demonstrating Green-e Equivalency

Projects not using Green-e certified products must demonstrate equivalency to the Green-e Energy standard requirements in the following areas:. To be considered equivalent to Green-e Energy, the EACs retired on behalf of the LEED project must be

- Certified under an eco-label or similar program developed by an independent organization with transparent accounting process and standards in place
- Certified under an eco-label developed by an independent organization with transparent accounting processes and standards
- Eligible renewable energy source (see Green-e Framework for Renewable Energy Certification, Section IIIA, "Renewable Resource Types", and additional regional requirements as applicable, i.e. Appendix D: Green -e Renewable Energy Standard for Canada and the United States, Section II (Eligible Sources of Supply)
- Renewable resources are from projects that have come online within the last fifteen years
- Verifiable chain of custody
- Mechanism to prevent double-counting

The executed contract must specify the purchasing goals consistent with the Green-e equivalency requirements and be valid for the duration indicated in the credit documentation.

Captured Bio-methane

Biogas from landfills, sewage treatment plants, and farm methane capture that is captured off-site and delivered to the LEED project for combustion on-site is considered "captured bio-methane" in LEED v4.1. Captured bio-methane may be used to offset fuel energy usage and qualify as Tier 3 off-site renewable energy provided that the biogas contract meets the minimum requirements for contract length. For procurement of captured bio-methane where a ten-year contract is not available, procurement shall be prorated based on the contract term length.

Retirement of environmental attributes of renewable gas is substantiated through certificates specific to gas production and use. The LEED project owner shall contract for renewable gas and demonstrate that the environmental attribute certificates are retained, owned, or retired on behalf of the LEED project.

Documenting Savings for EAp Minimum Energy Performance and EAc Optimize Energy Performance

Refer to EAp Minimum Energy Performance, Further Explanation, Applying Renewable Energy Savings.

Rating System Variations

Core and Shell

CS projects shall determine total site energy use in the same manner as other BD+C projects, including the total site energy associated with the base building and future tenant fit outs (see Further Explanation, Total Site Energy Use).

Note:
The total site energy reference is aligned with EAc Renewable Energy Production from previous versions of LEED, which evaluates the Core and Shell renewable energy percentage based on total building energy use. The credit does not include provisions for calculating a lower "core and shell building energy" based on Core and shell floor area as was previously allowed in EAc Green Power and Carbon Offsets.

Project Type Variations

District Energy Systems (DES)

The method for documenting EA Renewable Energy credit must be consistent with the method for documenting EAp Minimum Energy Performance. If the project is using a prescriptive path to document EAp Minimum Energy Performance, OR applies ASHRAE 90.1 Appendix G to model district energy using identical purchased energy rates (i.e. $/unit of delivered energy) and Greenhouse gas emissions factors (i.e. lb CO2eq emissions per unit of delivered district energy) in the Baseline and Proposed case for EAp Minimum Energy Performance, renewable electricity and EACs may be applied to offset the site energy use of the district energy (e.g. district steam, district hot water, district chilled water) in EAc Renewable Energy.

If the project documents a performance improvement for the district energy system within EAp Minimum Energy Performance using the LEED District Energy Guidance Path 2 or 3, the modeled results in the EAp Minimum Energy Performance will indicate the estimated natural gas, electricity, and/or other fuels associated with the district energy generation; and district energy will not be reported as an energy source in the modeled results. In this case, renewable electricity and EACs may only be applied to electricity (either in the building or associated with the district energy system) and captured bio-methane may only be applied to the fossil fuel or other fuel used in the building or associated with the district energy systems.

Required Documentation

Documentation
LEED v4.1 Renewable Energy Calculator
Renewable system rated capacity
Calculations to determine energy generated
Confirmation of renewable attribute ownership (for on-site systems owned by building owner) or contract indicating duration and renewable attribute ownership (for on-site systems owned by a third party)
Contract indicating percentage ownership, lease, or allocation of Tier 2 off-site renewable system
Purchase letter or contract of commitment showing renewable energy for targeted point threshold
Eco-label documentation showing Green-e Energy equivalency if not Green-e certified

Exemplary Performance

Tier 1: renewable energy generation meets or exceeds 25% of total site energy use.

Tier 2: renewable energy procurement meets or exceeds 60% of total site energy use.

Exemplary performance is not available for Tier 3 renewable energy procurement.

Connection to Ongoing Building Performance

- LEED O+M EA credit Energy Performance: Investments in renewable energy throughout the building life cycle can help reduce building greenhouse gas emissions and improve the building's energy performance score, increase market demand for renewables, and support the growth and financial feasibility of new renewable energy projects.